

1. Data Cleaning Strategy:

I. Handling Missing Values

Books Dataset:

- Drop rows missing essential info like Book-Title, Book-Author, or Publisher.
- Convert Year-Of-Publication to numeric.
- Converted invalid Year-Of-Publication entries (e.g., text) to NaN.
- Impute Year-Of-Publication using:
 - a. Replace with median year of the dataset if ISBN exists multiple times with valid years.
 - b. Mode of year column, if ISBN is unique.

Users Dataset:

- Fill missing Location with "Unknown".
- Fill missing Age with median/Mode
- drop if Age outside valid range (e.g., 5–100).

Ratings Dataset:

- Drop rows with missing User-ID or ISBN .

II. Removing Duplicates

Books dataset :

- Keep only one row per ISBN if metadata is identical.

Users dataset:

- Ensure User-ID is unique.
- Remove exact duplicate user records.

Ratings dataset:

- Remove rows with the same User-ID, ISBN, and Book-Rating.

III. Standardizing Data Formats

i. Text Fields (titles, authors, publishers, Location):

- Strip whitespaces.
- Convert all text fields (Book-Title, Book-Author, Publisher, Location) to lowercase.
- Normalize publisher/author spellings.
- Clean special characters (_ , * , -) in Location.

ii. Date Fields:

- Convert Year-Of-Publication to integer and apply standardization.
- Remove or fix non-year values (e.g., text in place of year).

iii. User Age:

- Set valid age range (e.g., keep only $5 \leq \text{Age} \leq 100$).

IV. Fixing Inconsistencies

- Replace incorrect Year-Of-Publication (e.g., < 1800 or > 2025) with Median or most frequent value for that ISBN.
- Remove invalid Publisher values (e.g., numeric-only names).
- Normalize Book-Rating: removing implicit ratings (i.e., 0) if not useful for modeling.

V. Outlier Detection

- Draw boxplot to detect outliers

a. Age (Users)

- Apply the Interquartile Range (IQR) method:
- Removed users with ages beyond $1.5 \times \text{IQR}$ from Q1 and Q3.
- Helps to eliminate improbable outliers like 0, 200, etc.

b. Year-Of-Publication (Books)

- Apply IQR filtering after removing invalid years (e.g., 0, 9999).
- Helped in identifying extremely old or wrongly entered publication years.

VI. General Cleaning Practices

- Use `.dropna()` for rows with critical null values.
- Use `.fillna()` for columns where imputation makes sense (e.g., age).

- Apply `.drop_duplicates()` to clean repeated data entries.
- Standardized ISBN as string format; validated length (10 or 13 characters).

2. Setting Up a Git Repository:

Repository Structure:

Bookify.com/

